



1.05 Burping the Vacuum Truck

1. Purpose

The purpose of the Standard Operating Procedure is to identify the hazards and define the steps in burping the vacuum trucks .

2. Scope

This procedure is intended for use by the Silverpoint employees .

3. Hazard Identification

- Physical Stress .
- Personal Injury .
- Equipment Damage .
- Fall from Height – Equipment /Personal .
- Pressurized Liquids .
- Pinch Points
- Ergonomics .
- Lifting , Twisting , Pushing and Pulling .
- Valves Partially Open .
- Plugged Spread Valve
- Product Spill
- Equipment Failure

4. EHS Requirements

- FR Coveralls .
- Steel Toed Boots .
- Safety Glasses .
- Rubber Gloves .
- Hard Hat .
- Hearing Protection .



- Redesign Pit with Warning Signs .
- Valves Properly Labeled .
- Check Valve Lever Position before Removing Cap, Remove Slowly Out of Line of Fire.
- Correct Lifting Training .
- PMI .

5. References

- Silverpoint Corporate Codes of Practice
- Alberta OH&S
- Task List
- MED HAT THA 1.05

6. Procedure

Burping should always be done with the truck stopped and parked on as level ground as possible. Burping the Vacuum Truck places a lot of strain on the truck frame, hydraulic hoist, truck suspension and the rear tank hinges, by following this procedure you can limit the wear and tear on the vacuum truck while burping.

6.1. The purpose of this procedure is to suspend all the solids or heavier debris throughout the tank by mixing the shale and liquids in the tank and to flush out the tank without having to open the back door. This procedure is also used to clear the 6" spread valve of shale before land spreading . It is necessary to raise the tank for burping to be effective . **Never raise the tank beyond the second stage of the hydraulic ram during any point of this procedure.**

6.1.1. If the 6-inch spread valve is plugged with sand or shale and using the agitator with full vacuum will not remove the material blocking it, it may be necessary to use a shovel handle , or the tank door lock out bar to loosen the material blocking the valve. Make sure you have all your PPE on before starting to unplug 6 inch spread valve, keep your hands away from the valve, keep your hands away from any possible pinch points , once the material breaks free there will be full vacuum at the valve .

6.1.2. You must ensure the entire vacuum system has vacuum before attempting to unblock spread valve, not just the system piping after the shut off ball. Do not attempt this if the tank is balled off, **if there is pressure on the tank you end up with fluid coming out, not air going in.**

6.1.3. Put the vacuum /pressure valve on vacuum and engage the Hybon , watch vacuum gauge to make sure the tank is building vacuum , not just the vacuum system piping .

6.1.4. Check that the entire system has vacuum by slowly opening the 4-inch load valve and make sure that air is



flowing into the tank.

- 6.1.5. Once you are sure that the system has vacuum, open the 6-inch spread valve and use the shovel handle or tank door lock out bar to loosen the material blocking the 6 inch Valve.
- 6.1.6. Work at the material slowly until a hole has opened up in the material and air can flow into the valve. **Do not pound on the valve.**
- 6.2. Using the Gallop agitator and the tank vibrator during this procedure can greatly increase the effectiveness of burping, the agitator will help stir the solids that are loosened from the bottom of the tank by burping and vibrating can help move the solids toward the back of the tank where they will be in the flow of the air going into the tank.
- 6.3. When unloading at a pit, sump or cement ring; place the truck on vacuum and engage the pump, idle up the vacuum truck to its proper operating speed for that unit (RPMS), **add 6-inch hose if unloading into cement ring**, then open the spread valve. This must be done quickly to ensure that the tank does not build too much vacuum.
- 6.4. **Too much vacuum would cause the truck to buck violently when opening the valve making the area dangerous to you and others and possibly damaging the truck.**
- 6.5. Depending on the amount of solids in the tank you may need to burp the truck for 5 to 10 minutes, when you are satisfied that you have loosened all the solids, disengage pump and neutralize the vacuum in the tank, letting the load flow out of the tank freely.
- 6.6. Burping the tank can be done in the spread field as well, but you will have to spread part of your load first so there is room in the tank, that way you don't ball off. Place the truck on vacuum and engage the pump, idle up the vacuum truck to its proper operating speed for that unit (RPMS) and then let it build a little vacuum. Then open the spread valve and it will start to burp and then close valve quickly, so you do not make a mess on the ground. Do it a few times if you have a lot of shale in vacuum truck on your last load. Be sure there is enough room in the tank to allow for the pocket of air to travel through the load without causing you to ball off. If it were to ball off, it would cause the mud to drain out of the 6-inch spread valve. As extra room is gained in the tank it will allow for the load to mix more thoroughly.
- 6.7. For maximum results let the vacuum truck burp for a couple of minutes at a time when at the cement pit or ring. Performing this procedure properly may save you from entering a confined space to clean out the tank.
- 6.8. When this procedure is done, there is a chance that some of the product might spill on the ground, if you are in the spread field make sure to check for spills. Clean up the spill if one occurs. **If you do have a spill, clean it up and notify your supervisor or technician right away, regardless of the size of the spill.**

Remember: if you are in doubt about what to do, call your supervisor